Harshini Nujella

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Education

University of California, Los Angeles (UCLA) — Master of Engineering, Data Science *Expected: August 2027*
University of California, Merced — Bachelor of Science, Computer Science & Engineering *Graduated: December 2025*

Technical Skills

Programming Languages: Python, C++, C, HTML/CSS, Java, SQL, JavaScript/TypeScript
AI/ML: PyTorch, TensorFlow, scikit-learn, Neural Networks, Hugging Face Transformers, OpenCV, AI Agents, LLMs, RAG, LangChain
Frameworks & Tools: AWS, Flask, PostgreSQL, Docker, Kubernetes, Azure, Pandas, Git, REST API, Unix/Linux

Experience

Undergraduate AI/ML Research Engineer | AI Systems and Education Lab *May 2025 – Aug 2026*

- Piloted SmartGrader website in an Intro to Python course, deployed on **Microsoft Azure** and integrated with the student portal via **LTI 1.3/OIDC** authentication and **SQLite** database, eliminating 100% of manual grading for 60 students, saving the team 6 hours/week.
- Developed an adaptive feedback engine using **GPT-4, agentic AI**, and **rule-augmented NLP**, with an error bank of 20 categories discovered by TF-IDF and DBSCAN clustering, to translate stack traces and coding errors into personalized guidance, improving student comprehension by **60%**.
- Authored 2 papers for the ASEE Conference and UCM URJ; selected to present at UC Merced Research Week 2026.

Machine Vision Intern | Owligator *Jan 2025 – Aug 2025*

- Reduced manual inventory tracking by 80% by leading E2E development of an AI wine label extraction system; built a full-stack web app (**React, Flask, Firebase**) and trained a custom **Roboflow** bottle detection model.
- Engineered a computer vision pipeline (**YOLOv8, SAM, OpenCV, OCR, OpenAI API**) achieving **95% accuracy** extracting label data from 10+ feet away in under 30 seconds.
- Created a comprehensive testing suite using **Jest** unit tests, **pytest** API integration tests, and load/stress testing to validate system reliability; diagnosed and resolved a critical race condition causing inconsistent inventory updates.

Cloud Technologies & AI/ML Intern | AI Squared *May 2024 – Aug 2024*

- Built a Python chatbot (**HuggingFace, LangChain, Llama 2**) enabling non-technical users to manage **SQL** databases through text, eliminating manual SQL command entry entirely.
- Authored a concept note for **Lean AI 2.0**, an **MLOps** platform, by analyzing data transmission and model deployment workflows; presented to stakeholders to align engineering and business teams.

Project

Spot It! Game Automation with Machine Vision | UC Merced *March 2025*

- Trained a **YOLOv8** model on a self-annotated dataset (70/20/10 train/val/test split), achieving **97.9% true positive rate**.
- Designed a **confidence-based matching algorithm** to resolve multiple detections using the max-confidence per region.
- Detected overfitting through **hyperparameter ablation**, and reduced it using **data augmentation** (color, noise & cropping).

Full-Stack IoT Sensor Analytics Platform | UC Merced *September 2024 – December 2024*

- Deployed an environmental monitoring system using **Raspberry Pi**, integrating real-time data from weather, water quality, and soil moisture sensors.
- Built a full-stack data pipeline with **RESTful APIs** and **Pandas** for automated data cleaning, transformation, and classification processing of 1k+ sensor readings.
- Designed **SQL schemas** and interactive dashboards with **D3.js** to support real-time visualization and statistical reporting.

Awards

Outstanding Student - School of Engineering, Class of Fall 2025 *2025*
Director's Choice Award, DAHacks *2024*
Winner, Blackstone LaunchPad – multi-campus entrepreneurship competition, Founder of CatCar (500+ users) *2023*
Winner, De Anza Business and Finance Village Competition *2023*